

Employee Table Schema with Data Integrity Rules

MySQL Assignment 1 — DDL Commands & Constraints

Project Details	
Created By	Sharu Latha B
Course	Data Analytics (DA) — Module 3
Tool	MySQL Workbench 8.0
Script	Employee_DB_OneSheet.sql

Part A — DDL Commands (CREATE, ALTER, RENAME, TRUNCATE, DROP)

Task 1 — Database & Table Creation (CREATE)

Create the project database, then Departments, Location, and Employees tables with foreign-key relationships.

Limit to 1000 rows ▾

1

DROP DATABASE IF EXISTS

employee

✓

Query OK, 3 rows affected (0.035 sec)

Limit to 1000 rows ▾

3

CREATE DATABASE

employee

✓

Query OK, 1 row affected (0.000 sec)

Limit to 1000 rows ▾

5

CREATE TABLE

Departments (

6

department_id

INT PRIMARY KEY,

7

department_name

VARCHAR(100)

8

)

✓

Query OK, 0 rows affected (0.006 sec)

Limit to 1000 rows ▾

10

CREATE TABLE

Location (

11

location_id

INT PRIMARY KEY,

12

location_name

VARCHAR(100)

13

)

✓

Query OK, 0 rows affected (0.008 sec)

Task 2 — Table Alteration (ALTER)

Add email, widen Designation, drop Age, and rename Hire_date → date_of_joining.

Limit to 1000 rows ▾

15

CREATE TABLE

Employees (

16

employee_id

INT PRIMARY KEY,

17

Employee_name

VARCHAR(50),

18

Gender

CHAR(1),

19

Age

INT,

20

Hire_date

DATE,

21

Designation

VARCHAR(50),

22

Salary

DECIMAL(10,2),

23

department_id

INT,

24

location_id

INT,

25

FOREIGN KEY (department_id)

REFERENCES Departments(department_id),

26

FOREIGN KEY (location_id)

REFERENCES Location(location_id)

27

)

✓

Query OK, 0 rows affected (0.010 sec)

Limit to 1000 rows ▾

29

ALTER TABLE

Employees

ADD COLUMN

email

VARCHAR(100)

✓

Query OK, 0 rows affected (0.011 sec) Records: 0 Duplicates: 0 Warnings: 0

Limit to 1000 rows ▾

31

ALTER TABLE

Employees

MODIFY COLUMN

Designation

VARCHAR(150)

✓

Query OK, 0 rows affected (0.020 sec) Records: 0 Duplicates: 0 Warnings: 0

Limit to 1000 rows ▾

33

ALTER TABLE

Employees

DROP COLUMN

Age

✓

Query OK, 0 rows affected (0.013 sec) Records: 0 Duplicates: 0 Warnings: 0

Task 3 — Table Renaming (RENAME)

Departments → Departments_Info, Location → Locations.

Limit to 1000 rows ▾

35 • **ALTER TABLE** Employees **RENAME COLUMN** Hire_date **TO** date_of_joining

✔ Query OK, 0 rows affected (0.012 sec) Records: 0 Duplicates: 0 Warnings: 0

Limit to 1000 rows ▾

37 • **RENAME TABLE** Departments **TO** Departments_Info

✔ Query OK, 0 rows affected (0.005 sec)

Task 4 — Table Truncation (TRUNCATE)

Clear all rows from the Employees table.

Limit to 1000 rows ▾

39 • **RENAME TABLE** Location **TO** Locations

✔ Query OK, 0 rows affected (0.006 sec)

Task 5 — Database & Table Dropping (DROP)

Drop the Employees table, then the database.

Limit to 1000 rows ▾

41 • **TRUNCATE TABLE** Employees

✔ Query OK, 0 rows affected (0.007 sec)

Limit to 1000 rows ▾

43 • **DROP TABLE** Employees

✔ Query OK, 0 rows affected (0.006 sec)

Part B — Constraints (Recreate Schema)

Task 1 — Database Recreation

Drop the database if it exists, then recreate it.

Limit to 1000 rows ▼

1

DROP DATABASE

employee

✓ Query OK, 2 rows affected (0.011 sec)

Limit to 1000 rows ▼

3

DROP DATABASE IF EXISTS

employee

✓ Query OK, 0 rows affected, 1 warning (0.000 sec)

Task 2 — Departments Table

department_id as PRIMARY KEY; department_name NOT NULL + UNIQUE.

Limit to 1000 rows ▼

5

CREATE DATABASE

employee

✓ Query OK, 1 row affected (0.000 sec)

Task 3 — Locations Table

location_id as AUTO_INCREMENT PRIMARY KEY; location_name NOT NULL + UNIQUE.

Limit to 1000 rows ▼

7

CREATE TABLE

Departments (

8

department_id

INT PRIMARY KEY,

9

department_name

VARCHAR(100) NOT NULL UNIQUE

10

)

✓ Query OK, 0 rows affected (0.007 sec)

Task 4 — Employees Table

PRIMARY KEY, NOT NULL name, CHECK on Gender/Age, DEFAULT hire date, UNIQUE email, FOREIGN KEYS to Departments & Locations.

Limit to 1000 rows ▼

12

CREATE TABLE

Locations (

13

location_id

INT AUTO_INCREMENT PRIMARY KEY,

14

location_name

VARCHAR(100) NOT NULL UNIQUE

15

)

✓ Query OK, 0 rows affected (0.008 sec)

Limit to 1000 rows ▼

17

CREATE TABLE

Employees (

18

employee_id

INT PRIMARY KEY,

19

Employee_name

VARCHAR(50) NOT NULL,

20

Gender

CHAR(1) CHECK (Gender IN ('M', 'F')),

21

Age

INT CHECK (Age >= 18),

22

Hire_date

DATE DEFAULT (CURRENT_DATE),

23

Designation

VARCHAR(150),

24

Salary

DECIMAL(10,2),

25

email

VARCHAR(100) UNIQUE,

26

department_id

INT,

27

location_id

INT,

28

CONSTRAINT fk_employee_department

29

FOREIGN KEY (department_id) REFERENCES Departments(department_id),

30

CONSTRAINT fk_employee_location

31

FOREIGN KEY (location_id) REFERENCES Locations(location_id)

32

)

✓ Query OK, 0 rows affected (0.011 sec)

Constraint Enforcement — Proof of Validation

Test 1 — Invalid Gender

CHECK (Gender IN ('M','F')) should reject 'X'.

Limit to 1000 rows ▼

34

INSERT INTO

Employees (employee_id, Employee_name, Gender, Age) VALUES (999, 'ConstraintTest', 'X', 20)

✓

Test 2 — Underage Employeee

CHECK (Age >= 18) should reject age 15.

Limit to 1000 rows

36 [INSERT INTO](#) Employees (employee_id, Employee_name, Gender, Age) [VALUES](#) (998, 'ConstraintTest2', 'M', 15)

Test 3 — Duplicate department_id

PRIMARY KEY should reject a repeated id.

Limit to 1000 rows

38 [INSERT INTO](#) Departments (department_id, department_name) [VALUES](#) (1, 'Duplicate')

Final Schema Verification

Limit to 1000 rows

40 [SHOW TABLES](#)

Result Grid

Filter Rows:

Edit: [🔍](#) [+](#) [×](#) Export/Import: [📄](#)

	Tables_in_employee
▶	Departments
▶	Employeees
▶	Locations

3 rows in set (0.000 sec)

Limit to 1000 rows

42 [DESCRIBE](#) Employees

Result Grid

Filter Rows:

Edit: [🔍](#) [+](#) [×](#) Export/Import: [📄](#)

	Field	Type	Null	Key	Default	Extra
▶	employee_id	int(11)	NO	PRI	NULL	
▶	Employee_name	varchar(50)	NO		NULL	
▶	Gender	char(1)	YES		NULL	
▶	Age	int(11)	YES		NULL	
▶	Hire_date	date	YES		curdate()	
▶	Designation	varchar(150)	YES		NULL	
▶	Salary	decimal(10,2)	YES		NULL	
▶	email	varchar(100)	YES	UNI	NULL	
▶	department_id	int(11)	YES	MUL	NULL	
▶	location_id	int(11)	YES	MUL	NULL	

10 rows in set (0.000 sec)

Data Exploration — Sample Data & Joins

Sample Data Load

Populate Departments, Locations, and Employees with test records.

Limit to 1000 rows ▼

1

•

INSERT INTO Departments (department_id, department_name) VALUES

2

•

(1, 'Human Resources'),

3

•

(2, 'Engineering'),

4

•

(3, 'Sales'),

5

•

(4, 'Finance')

Query OK, 4 rows affected (0.001 sec) Records: 4 Duplicates: 0 Warnings: 0

Join 1 — INNER JOIN

Employees matched with their department and location names.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

SCHEMAS

Filter objects

analysis_db
Tables
Views
Stored Procedures
Functions
employee
Tables
departments
employees
locations
Views
Stored Procedures
Functions
sakila

Administration Schemas

Information

Table: departments
Columns:
department_id int PK
department_name varchar(100)

Query 1 Employee_DB_OneSheet (1) x

Limit to 1000 rows

8 USE employee;
9
10 CREATE TABLE Departments (
11 department_id INT PRIMARY KEY,
12 department_name VARCHAR(100)
13);
14
15 CREATE TABLE Location (
16 location_id INT PRIMARY KEY,
17 location_name VARCHAR(100)
18);

Result Grid

Filter Rows: Export: Wrap Cell Content: I

employee_id	Employee_name	department_name	location_name	Designation	Salary	email
101	Riya Sharma	Engineering	Bengaluru	Software Engineer	65000.00	riya.sharma@company.com
104	Vikram Nair	Finance	Bengaluru	Finance Director	95000.00	vikram.nair@company.com
105	Sneha Iyer	Engineering	Bengaluru	Software Engineer	68000.00	sneha.iyer@company.com
102	Arjun Mehta	Human Resources	Delhi	HR Manager	72000.00	arjun.mehta@company.com
106	Rohan Gupta	Sales	Delhi	Sales Executive	42000.00	rohan.gupta@company.com
108	Karan Malhotra	Engineering	Delhi	Software Engineer	70000.00	karan.malhotra@company.com
103	Kavya Reddy	Sales	Mumbai	Sales Executive	40000.00	kavya.reddy@company.com
107	Ananya Das	Human Resources	Mumbai	HR Executive	50000.00	ananya.das@company.com

EMPLOYEE x department Location Deptheadcount dept salary total salary gender dept avg salary

Output

Action Output

Time Action Message Duration / Fetch

Join 2 — LEFT JOIN

All departments, including any with zero employees.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

SCHEMAS

Filter objects

analysis_db
Tables
Views
Stored Procedures
Functions
employee
Tables
departments
Columns
department_id
department_name
Indexes
Foreign Keys
Triggers
sakila

Administration Schemas

Information

Table: departments
Columns:
department_id int PK
department_name varchar(100)

Query 1 Employee_DB_OneSheet (1) x

Limit to 1000 rows

14
15 CREATE TABLE Location (
16 location_id INT PRIMARY KEY,
17 location_name VARCHAR(100)
18);
19
20 CREATE TABLE Employees (
21 employee_id INT PRIMARY KEY,
22 Employee_name VARCHAR(50),
23 Gender CHAR(1)
24);

Result Grid

Filter Rows: Export: Wrap Cell Content: I

department_id	department_name	employee_id	Employee_name
2	Engineering	101	Riya Sharma
2	Engineering	105	Sneha Iyer
2	Engineering	108	Karan Malhotra
4	Finance	104	Vikram Nair
1	Human Resources	102	Arjun Mehta
1	Human Resources	107	Ananya Das
3	Sales	103	Kavya Reddy
3	Sales	106	Rohan Gupta

EMPLOYEE department x Location Deptheadcount dept salary total salary gender dept avg salary

Output

Action Output

Time Action Message Duration / Fetch

Join 3 — LEFT JOIN

All locations, including any with zero employees.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

Views

Stored Procedures

Functions

employee

Tables

departments

employees

locations

Columns

Indexes

Foreign Keys

Triggers

Views

Stored Procedures

Administration

Schemas

Information

Table: locations

Columns:

location_id int AI PK

location_name varchar(100)

Query 1

Employee_DB_OneSheet (1)

Limit to 1000 rows

14

15 CREATE TABLE Location (

16 location_id INT PRIMARY KEY,

17 location_name VARCHAR(100)

18);

19

20 CREATE TABLE Employees (

21 employee_id INT PRIMARY KEY,

22 Employee_name VARCHAR(50),

23

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

1

location_id location_name employee_id Employee_name

1 Bengaluru 101 Riya Sharma

1 Bengaluru 104 Vikram Nair

1 Bengaluru 105 Sneha Iyer

3 Delhi 102 Arjun Mehta

3 Delhi 106 Rohan Gupta

3 Delhi 108 Karan Malhotra

2 Mumbai 103 Kavya Reddy

2 Mumbai 107 Ananya Das

EMPLOYEE department Location x Deptheadcount deptsalary totalsalary gender deptavg salary

Output

Action Output

Time Action Message Duration / Fetch

Result Grid

Form Editor

Five Analytical Queries

Q1

Headcount and average salary per department.

The screenshot shows the MySQL Workbench interface. The 'Schemas' pane on the left displays the 'employee' database with tables 'departments', 'employees', and 'locations'. The 'Query 1' editor shows the following SQL code:

```
14
15 CREATE TABLE Location (
16     location_id INT PRIMARY KEY,
17     location_name VARCHAR(100)
18 );
19
20 CREATE TABLE Employees (
21     employee_id INT PRIMARY KEY,
22     Employee_name VARCHAR(50),
23     Gender CHAR(1)
```

The 'Result Grid' shows the output of the query:

department_name	headcount	avg_salary
Finance	1	95000.00
Engineering	3	67666.67
Human Resources	2	61000.00
Sales	2	41000.00

The 'Table: employees' information pane on the left lists the columns and their data types:

- employee_id: int PK
- Employee_name: varchar(50)
- Gender: char(1)
- Age: int
- Hire_date: date
- Designation: varchar(150)
- Salary: decimal(10,2)
- email: varchar(100)

Q2

Highest-paid employee in each department.

The screenshot shows the MySQL Workbench interface. The 'Schemas' pane on the left displays the 'employee' database with tables 'departments', 'employees', and 'locations'. The 'Query 1' editor shows the following SQL code:

```
14
15 CREATE TABLE Location (
16     location_id INT PRIMARY KEY,
17     location_name VARCHAR(100)
18 );
19
20 CREATE TABLE Employees (
21     employee_id INT PRIMARY KEY,
22     Employee_name VARCHAR(50),
23     Gender CHAR(1)
```

The 'Result Grid' shows the output of the query:

department_name	Employee_name	Salary
Human Resources	Arjun Mehta	72000.00
Finance	Vikram Nair	95000.00
Sales	Rohan Gupta	42000.00
Engineering	Karan Malhotra	70000.00

The 'Table: employees' information pane on the left lists the columns and their data types:

- employee_id: int PK
- Employee_name: varchar(50)
- Gender: char(1)
- Age: int
- Hire_date: date
- Designation: varchar(150)
- Salary: decimal(10,2)
- email: varchar(100)

Q3

Employee count and total salary cost per location.

MySQL Workbench interface showing the creation of tables and the resulting data.

Query 1: Employee_DB_OneSheet (1)

```
14
15 CREATE TABLE Location (
16     location_id INT PRIMARY KEY,
17     location_name VARCHAR(100)
18 );
19
20 CREATE TABLE Employees (
21     employee_id INT PRIMARY KEY,
22     Employee_name VARCHAR(50),
23     Gender VARCHAR(10),
24     Age INT,
25     Hire_date DATE,
26     Designation VARCHAR(150),
27     Salary decimal(10,2),
28     email VARCHAR(100)
29 );
```

Result Grid:

location_name	num_employees	total_salary_cost
Bengaluru	3	228000.00
Delhi	3	184000.00
Mumbai	2	90000.00

Table: employees

Columns:

- employee_id: int PK
- Employee_name: varchar(50)
- Gender: char(1)
- Age: int
- Hire_date: date
- Designation: varchar(150)
- Salary: decimal(10,2)
- email: varchar(100)

Q4

Gender distribution and average age by department.

MySQL Workbench interface showing the creation of tables and the resulting data.

Query 1: Employee_DB_OneSheet (1)

```
14
15 CREATE TABLE Location (
16     location_id INT PRIMARY KEY,
17     location_name VARCHAR(100)
18 );
19
20 CREATE TABLE Employees (
21     employee_id INT PRIMARY KEY,
22     Employee_name VARCHAR(50),
23     Gender VARCHAR(10),
24     Age INT,
25     Hire_date DATE,
26     Designation VARCHAR(150),
27     Salary decimal(10,2),
28     email VARCHAR(100)
29 );
```

Result Grid:

department_name	Gender	num_employees	avg_age
Engineering	F	2	29.0
Engineering	M	1	29.0
Finance	M	1	45.0
Human Resources	F	1	39.0
Human Resources	M	1	34.0
Sales	F	1	22.0
Sales	M	1	26.0

Table: employees

Columns:

- employee_id: int PK
- Employee_name: varchar(50)
- Gender: char(1)
- Age: int
- Hire_date: date
- Designation: varchar(150)
- Salary: decimal(10,2)
- email: varchar(100)

Q5

Departments where average salary exceeds the company-wide average.

MySQL Workbench

Local instance MySQL80

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

SCHEMAS

Filter objects

employee

Tables

departments

Columns

Indexes

Foreign Keys

Triggers

employees

Columns

Indexes

Foreign Keys

Triggers

locations

Columns

AdministrationSchemas

Information

Query 1Employee_DB_OneSheet (1)

Limit to 1000 rows

14

15CREATE TABLE Location (

16location_id INT PRIMARY KEY,

17location_name VARCHAR(100)

18);

19

20CREATE TABLE Employees (

21employee_id INT PRIMARY KEY,

22Employee_name VARCHAR(50),

23Gender CHAR(1)

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Read Only

department_name	dept_avg_salary
Engineering	67666.67
Finance	95000.00

Table: employees

Columns:

employee_idint PK

Employee_namevarchar(50)

Genderchar(1)

Ageint

Hire_date date

Designation varchar(150)

Salary decimal(10,2)

email varchar(100)

EMPLOYEE

department

Location

Deptheadcount

deptsalary

total salary

gender

dept avg salary

Output

Action Output

#

Time

Action

Message

Duration / Fetch